



KRISHNAMOORTHY S

EMBEDDEDENGINEER|C| BE -ECE



MADURAI

(+91) 6369948483

krishnamoorthys1408@gmail.com

Enthusiastic and results-driven Embedded Systems Engineer with 1 years of professional experience in firmware development, microcontroller programming, and IoT applications. Skilled in working with 8051, PIC, ARM, ESP32, and Arduino microcontrollers, with strong expertise in communication protocols (UART, SPI, I2C, CAN, MQTT, BLE) and hardware integration. Seeking to contribute technical expertise and innovative problem-solving skills to develop robust embedded solutions in a dynamic organization.

EDUCATION :

B.E. ECE - Velammal college of Engineering and Technology, Madurai 2021 –2025	CGPA: 8.2
HSC - TVS Higher Secondary school, Madurai 2021	Percentage: 89.25%
SSLC - TVS Higher Secondary school, Madurai 2019	Percentage: 86.4%

WORK EXPERIENCE :

Embedded Engineer - Petal Automations, Madurai. May 2025 - Present

1. IOT LAB KIT

Team Size: 3

Role:Embedded Software Developer.

Responsibility: Developed firmware for ESP32 and Arduino boards.

Technology used: IOT, RS485, CAN, USB to TTL, RFID, Bluetooth, BLE, MQTT, UART, SPI and I2C communication Protocols.

ProjectOverview: Designed an IoT Learning Kit for students to explore interfacing of input/output devices and sensormodules with IoT platforms. The kit enables hands-on learning of multiple communication protocols and sensor integration, helping students understand real-time IoT applications and connectivity.

2. GYM AUTOMATION

Team Size: 3

Role:Embedded Software Developer.

Responsibility: Writing code in ESP32 and PIC.

Technology used: IOT, RFID, MQTT, UART and SPI communication Protocols.

ProjectOverview: Designed a Gym Automation System where RFID-based locker assignment is automated. Locker door and obstacle status are monitored using sensors, and real-time updates are shared via MQTT to a mobile app/dashboard for efficient locker management.

3. STEM Project

Team Size: 3

Role: Embedded Software Developer

Responsibility: Implemented one-wire communication protocol between Arduino and STC15 series Microcontrollers. Developed and tested firmware for reliable data transfer.

Technology Used: One-Wire Protocol, Arduino, STC15 Microcontrollers, UART

Project Overview: Designed a master-slave communication system using a single-wire interface for efficient command transfer and device control in educational applications.

ACADEMIC PROJECT :

1. Water Quality Monitoring System for Biofloc Aquaculture

Objective: Developed an innovative system to monitor and control water quality in biofloc aquaculture, enhancing operational efficiency and sustainability.

Technologies Used: Arduino, DS18B20, pH sensor, TCS34725, ESP01, Blynk app.

Outcome: Enhanced water quality management, reduced manual labor, and improved aquaculture productivity and sustainability.

INTERNSHIP :

Embedded Engineer Trainee – Kalanjiyam Technical Solutions Private Limited, Madurai, 2024

Successfully learned C programming, Arduino & ESP32 development, and IoT fundamentals and Worked with various sensors, LCDs, and GPIO to design automation projects.

PROFESSIONAL SKILL:

Embedded Systems: 8051, PIC, ARM, Arduino, ESP32

Programming Languages: C, C++, Embedded C

Communication Protocols: SPI, UART, I2C, CAN, RS485

Technologies : BLE, WiFi, MQTT

Hardware: Schematic Design